

Effective Stiffness Tensor for the Perfectly Matched Layer: A Transversely Isotropic Representation

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1 Form Invariance of Elastic Wave equation

The governing equation of elastic waves, Navier's equation, is written as

$$\text{div} \mathbf{C}[\text{grad} \mathbf{u}] + \rho \omega^2 \mathbf{u} = 0 \quad (1)$$

where $\mathbf{C}$ is the fourth-order elastic tensor, $\mathbf{u}$ is the wave displacement vector, ρ is the mass density, and ω is the angular frequency. As the core of transformation elastodynamics, the form invariance of Eq. (1) under coordinate transformation has been investigated by various methods. This phenomenon arises because displacement is an intrinsic physical quantity that is not tied to a specific coordinate system but is instead represented differently under coordinate transformations. Consequently, various methodologies have been developed depending on how the displacement vector is represented under coordinate changes, i.e., according to the choice of gauge (gauge freedom).

Among them, two methodological approaches are most commonly discussed. For the coordinate transformation from the virtual domain x^i to the physical domain X^I , let the Jacobian matrix be defined as

$$\lambda_I^i = \frac{\partial x^i}{\partial X^I}.$$

Under this definition, the first approach chooses the displacement gauge as λ_i^I , whereas the second approach adopts the identity transformation as the gauge, i.e., δ_i^I . Due to this choice of gauge, each approach is formulated in a different form.

In the first approach, where the gauge is chosen using the Jacobian matrix, Navier's equation no longer preserves its original form and is transformed into the Willis form, in which the stress tensor depends not only on the displacement but also on the momentum. In this case, the corresponding material properties are given as follows.

$$\begin{aligned} \tilde{C}^{IJKL} &= J^{-1} \Lambda_i^I \Lambda_j^J \Lambda_k^K \Lambda_l^L C^{ijkl} \\ \rho^{IJ} &= J^{-1} \Lambda_i^I \Lambda_j^J \delta^{ij} \rho \end{aligned} \quad (2)$$

where $J = \det(\Lambda_i^I)$ is the Jacobian. Here, the density becomes an anisotropic tensor, while the stiffness tensor C preserves both major and minor symmetries.

In the second approach, where the gauge is chosen as the identity transformation, the governing equation in the physical domain likewise takes the form of Navier's equation, and the corresponding material properties are given as follows.

$$\begin{aligned} \tilde{C}^{IJKL} &= J^{-1} \Lambda_i^I \delta_j^J \Lambda_k^K \delta_l^L C^{ijkl} \\ \tilde{\rho} &= J^{-1} \rho \end{aligned} \quad (3)$$

In this case, the density is still given in the form of a constant scaled by the inverse Jacobian, whereas the stiffness tensor C loses minor symmetry.[3]

Both approaches exhibit limitations, such as a change in the form of the governing equation during the transformation or a breakdown of symmetry in the principal material properties, and therefore cannot be regarded as fully satisfactory transformation schemes. However, from the standpoint of form invariance of the governing equation, the second approach can be considered to be more appropriate.

In particular, when the Jacobian matrix Λ takes a special form, the transformed stiffness tensor $\tilde{C}$ reduces, within the framework of micropolar elasticity, to a polar stiffness tensor with distinctive properties. This feature is particularly evident in the application of a perfectly matched layer (PML), since, when the PML is regarded as a coordinate transformation, the complex stretching is introduced independently along each coordinate axis. Consequently, the Jacobian matrix Λ assumes the special form of a diagonal matrix.

From this point onward, we consider the case in which a PML is applied in the x -direction and interpret the resulting medium as a transversely isotropic medium with respect to the x -axis. The transformed stiffness tensor is then regarded as the stiffness tensor of a transversely isotropic polar solid.

2 Transversely Isotropic Polar Stiffness Tensor Analysis

2.1 Decomposition of the displacement gradient: Transversely Isotropic Medium case

Before proceeding, we must first examine how the stiffness tensor of a transversely isotropic polar solid is represented in a general three-dimensional setting. Previous work by Nassar et al.[2] derived the representation of the stiffness tensor for an isotropic polar solid. In this paper, the displacement gradient is decomposed into three rotationally invariant subspaces under isotropic conditions. These subspaces are identified as spherical tensors, symmetric deviatoric tensors, and skew-symmetric tensors, and are expressed mathematically as follows. For displacement gradient $\mathbf{e}$,

$$\mathbf{e} = \frac{t}{3}\mathbf{I} + \tilde{\mathbf{e}} + \mathbf{W} \quad (4)$$

where $t = \text{tr}(\mathbf{e})$, $\mathbf{W} = \frac{\mathbf{e} - \mathbf{e}^T}{2}$, $\tilde{\mathbf{e}} = \frac{\mathbf{e} + \mathbf{e}^T}{2} - \frac{\text{tr}(\mathbf{e})}{3}\mathbf{I}$.

However, unlike an isotropic medium, a transversely isotropic medium exhibits only axial symmetry about a specific axis, rather than full rotational symmetry. This limitation means that directly applying the decomposition framework proposed in the aforementioned study is no longer sufficient. To address this, the decomposition must be further refined so that each constituent component remains rotationally invariant under the restricted symmetry condition imposed by a single preferred axis—that is, the irreducible decomposition must be carried out within the context of transverse isotropy.

Assuming the existence of symmetry about an axis $\mathbf{n}$, the three rotationally invariant tensors defined in the aforementioned study can be decomposed as follows. First, the spherical tensor $\mathbf{t}$ can be decomposed into the longitudinal stretch along the axis of symmetry and the areal dilation in the plane perpendicular to that axis. The skew-symmetric tensor $\mathbf{W}$ can be decomposed into the spin about the symmetry axis and the rigid body rotation of the volume element as a whole. Furthermore, the symmetric deviatoric tensor $\tilde{\mathbf{e}}$ can be decomposed into the in-plane

shear deformation and the shear deformation between the axial component and the in-plane component. This can be expressed mathematically as follows.

$$\mathbf{e} = \varepsilon_{nn} + \varepsilon_{\perp} + \mathbf{q} + E_{\perp} + \omega_{\parallel} \mathbf{n} + \omega_{\perp} \quad (5)$$

Denoting by $\mathbf{P}$ the projection matrix onto the plane perpendicular to the axis $\mathbf{n}$, each term can be computed as follows.

- **(Projection tensor):**

$$P = I - n \otimes n$$

- **(Axial Normal strain):**

$$\varepsilon_{nn} = n \cdot (\varepsilon n)$$

- **(In-plane volumetric strain):**

$$\varepsilon_{\perp} = \text{tr}(P\varepsilon)$$

- **(In-plane deviatoric strain):**

$$E_{\perp} = P\varepsilon P - \frac{\varepsilon_{\perp}}{2}P$$

- **(Out-of-plane shear strain):**

$$q = P\varepsilon n$$

- **(Rotation vector):**

$$\omega = \omega_{\parallel} n + \omega_{\perp}$$

Here, ω is adopted as the rotation vector because a skew-symmetric tensor $\mathbf{W}$ can be expressed as a vector representing the rotational component by means of the Levi-Civita symbol.

2.2 Deriving Transversely Isotropic Polar Solid Stiffness Tensor

We now proceed to compute the strain energy ψ through the quadratic forms of the aforementioned components, from which the stiffness tensor of a transversely isotropic polar medium is subsequently derived. To derive this, it is first necessary to determine which combinations of terms yield values that are invariant under rotations about the axis n . To this end, Schur's Lemma will be employed.

2.2.1 Schur's Lemma

Let G denote the symmetry group describing the rotational and symmetry transformations of a continuum (e.g., the three-dimensional orthogonal group $O(3)$ or the special orthogonal group $SO(3)$). It is assumed that the tensor space representing the deformation state of the medium of interest decomposes into two independent irreducible representations (irreps) V and W of the group G . The linear map (intertwining operator) $T : V \rightarrow W$ is then defined as the operator that maps the tensor state in the irreducible subspace V to the subspace W , thereby constructing the cross terms of the strain energy density.

Since the constitutive equations must satisfy the principle of material objectivity (frame-indifference), the linear map T is required to be equivariant with respect to the group action of G . That is, for any rotation $g \in G$ and tensor $x \in V$, the matrix T must commute with the rotation operator, which is expressed as follows.

$$T(g \cdot x) = g \cdot T(x) \quad (\forall g \in G, \forall x \in V) \quad (6)$$

For a linear map T satisfying the equivariance condition (G -equivariance) stated above, Schur's Lemma from the theory of group representations yields the following two inevitable conclusions based on the symmetry of the tensor space.

Lemma 1: Zero Map for Non-isomorphic Irreducible Representations

If V and W are non-isomorphic irreducible representations possessing distinct symmetry properties, then the only equivariant linear map connecting the two spaces is the zero map. Consequently, the coupling matrix reduces to $T = 0$, which physically implies that no cross terms in the strain energy can exist between deformation modes of differing symmetry.

Lemma 2: Scalar Multiplicity for Isomorphic Irreducible Representations

If V and W are isomorphic irreducible representations sharing identical symmetry properties ($V \cong W$), then any equivariant linear map connecting them must take the form of a scalar multiple of the identity operator I . Accordingly, the coupling matrix is of the form $T = \lambda I$, where λ is a real or complex scalar constant, and the resulting constant λ is defined as the intrinsic elastic stiffness coefficient of the material associated with the corresponding coupling mode.

That is, it follows from the irreducible decomposition in the transversely isotropic medium carried out above that only isomorphic components sharing identical symmetry properties and transformation behavior under rotation can contribute to the strain energy.

2.2.2 Quadratic form of deformation energy

From the foregoing conclusion, the symmetry of each component can be classified based on the weight m corresponding to the rotation angle in the $SO(2)$ plane.

First, the scalar quantities ε_{nn} , $\varepsilon_{\perp}$, $\omega_{\parallel}$ remain invariant under any rotation of the plane, corresponding to a weight of $m = 0$. For the rank-1 tensors $\mathbf{q}$ and $\omega_{\perp}$, if the rotation angle of the plane is θ , these quantities rotate by the same angle θ , corresponding to a weight of $m = 1$. In contrast, for the rank-2 tensor $E_{\perp}$, a rotation of the plane by θ induces a rotation of 2θ in the corresponding quantity, corresponding to a weight of $m = 2$. It can therefore be observed that, within the present irreducible decomposition, components of equal rank share identical symmetry properties with respect to the $SO(2)$ plane.

However, it cannot be concluded that all components of equal rank necessarily contribute to the strain energy through their mutual coupling. The most significant distinction lies in the mirror symmetry of each component. The rotational components $\omega_{\parallel}$, $\omega_{\perp}$ being axial vector quantities, remain unchanged under mirror reflection. In contrast, the remaining components — representing in-plane and out-of-plane shear, as well as tensile and compressive deformations — change sign under mirror reflection. Consequently, under the centrosymmetry condition, coupling between these two classes of components is precluded.

In summary, even within components of identical rank, coupling into the strain energy is permissible only for those terms that satisfy the centrosymmetry condition, which can be expressed as follows.

$$\begin{aligned}\psi = & \frac{1}{2}A\varepsilon_{nn}^2 + B\varepsilon_{nn}\varepsilon_{\perp} + \frac{1}{2}C\varepsilon_{\perp}^2 + DE_{\perp} : E_{\perp} + 2L\mathbf{q} \cdot \mathbf{q} \\ & + \frac{1}{2}\alpha_{\parallel}\omega_{\parallel}^2 + \frac{1}{2}\alpha_{\perp}\omega_{\perp} \cdot \omega_{\perp} + \gamma\omega_{\perp} \cdot (n \times q)\end{aligned}\quad (7)$$

Of particular importance is the last term. Contrary to what was stated earlier, it can be observed that q and $\omega_{\perp}$ are coupled to contribute to the strain energy. This is made possible by taking the cross product of q with the axial vector n , thereby reconciling the mirror symmetry properties of q and $\omega_{\perp}$. Furthermore, since such a transformation is not applicable to scalar quantities, no analogous coupling term exists among the rank-0 components.

2.2.3 Deriving Stiffness Tensor

We now proceed to derive the stiffness tensor from the quadratic form of the strain energy density established above. Under the assumption of linear elasticity, the following general relationship holds between the stiffness tensor and the deformation energy.

$$\frac{\partial^2 \psi}{\partial e_{ij} \partial e_{kl}} = C^{ijkl} \quad (8)$$

To make use of the above relation, the strain energy density ψ will first be expressed in index notation, after which it will be partially differentiated with respect to the displacement gradient e .

The strain components derived in Section 2.2.1 can be expressed in index notation as follows.

- (Projection tensor):

$$P^{ij} = \delta^{ij} - n^i n^j$$

- (Axial Normal strain):

$$\varepsilon_{nn} = n^i e_{ij} n^j$$

- (In-plane volumetric strain):

$$\varepsilon_{\perp} = P^{ij} e_{ij}$$

- (In-plane deviatoric strain):

$$E_{\perp}^{ij} = P^{im} e_{mn} P^{nj} - \frac{1}{2}(P^{mn} e_{mn}) P^{ij}$$

- (Out-of-plane shear strain):

$$q^i = P^{im} e_{mn} n^n$$

- (Rotation vector):

$$\omega^m = \epsilon^{mij} e_{ij}$$

Substituting these expressions into each energy term and rearranging into the form $\psi = \frac{1}{2}a^{ijkl}e_{ij}e_{kl}$, the following representation is obtained.

- **Term 1:** $\frac{1}{2}A\varepsilon_{nn}^2$

$$\frac{1}{2}A\varepsilon_{nn}^2 = \frac{1}{2}A(n^i n^j e_{ij})(n^k n^l e_{kl}) = \frac{1}{2}(An^i n^j n^k n^l)e_{ij}e_{kl}$$

- **Term 2:** $B\varepsilon_{nn}\varepsilon_{\perp}$

$$\begin{aligned} B\varepsilon_{nn}\varepsilon_{\perp} &= B(n^i n^j e_{ij})(P^{kl} e_{kl}) = Bn^i n^j P^{kl} e_{ij} e_{kl} \\ &= \frac{1}{2}B(n^i n^j P^{kl} + P^{ij} n^k n^l) e_{ij} e_{kl} \end{aligned}$$

- **Term 3:** $\frac{1}{2}C\varepsilon_{\perp}^2$

$$\frac{1}{2}C\varepsilon_{\perp}^2 = \frac{1}{2}C(P^{ij} e_{ij})(P^{kl} e_{kl}) = \frac{1}{2}CP^{ij}P^{kl}e_{ij}e_{kl}$$

- **Term 4:** $DE_{\perp} : E_{\perp}$

$$\begin{aligned} DE_{\perp}^{ij}E_{\perp}^{ij} &= D\left(P^{ik}e_{kl}P^{jl} - \frac{1}{2}P^{kl}e_{kl}P^{ij}\right)\left(P^{ki}e_{ij}P^{lj} - \frac{1}{2}P^{ij}e_{ij}P^{kl}\right) \\ &= D\left(P^{ik}P^{jl} - \frac{1}{2}P^{ij}P^{kl}\right)e_{ij}e_{kl} \\ &= \frac{1}{2}D\left(P^{ik}P^{jl} + P^{il}P^{jk} - P^{ij}P^{kl}\right)e_{ij}e_{kl} \end{aligned}$$

- **Term 5:** $2Lq \cdot q$

$$\begin{aligned} 2Lq^m q^m &= 2L(P^{mi}e_{ij}n^j)(P^{mk}e_{kl}n^l) = 2L(P^{ik}n^j n^l)e_{ij}e_{kl} \\ &= \frac{1}{2}L(P^{ik}n^j n^l + P^{il}n^j n^k + P^{jk}n^i n^l + P^{jl}n^i n^k) \end{aligned}$$

- **Term 6:** $\frac{1}{2}\alpha_{\parallel}\omega_{\parallel}^2$

$$\frac{1}{2}\alpha_{\parallel}\omega_{\parallel}^2 = \frac{1}{2}\alpha_{\parallel}(\omega^m n_m)(\omega^n n_n) = \frac{1}{2}\left(\frac{1}{4}\alpha_{\parallel}n_m n_n \epsilon^{mij}\epsilon^{nkl}\right)e_{ij}e_{kl}$$

- **Term 7:** $\frac{1}{2}\alpha_{\perp}\omega_{\perp} \cdot \omega_{\perp}$

$$\begin{aligned} \frac{1}{2}\alpha_{\perp}\omega_{\perp}^i \cdot \omega_{\perp}^i &= \frac{1}{2}(\alpha_{\perp}(P_m^i \omega^m)(P_n^i \omega^n)) = \frac{1}{2}(\alpha_{\perp}P_{mn}(\omega^m \omega^n)) \\ &= \frac{1}{2}\left(\frac{1}{4}\alpha_{\perp}P_{mn}\epsilon^{mij}\epsilon^{nkl}\right)e_{ij}e_{kl} \end{aligned}$$

- **Term 8:** $\gamma\omega_{\perp} \cdot (n \times q)$

$$\begin{aligned} \gamma\omega_{\perp}^i \cdot (n \times q)_i &= \gamma\left(\frac{1}{2}\epsilon^{kab}\omega_{ab}\right)(\epsilon_{kji}n^j q^i) = \frac{1}{2}\gamma(\delta_j^a \delta_k^b - \delta_k^a \delta_j^b)\omega_{ab}n^j q^k \\ &= \gamma\omega_{ij}n^j q^i = \gamma\omega_{ij}n^j(\delta^{ik} - n^i n^k)e_{kl}n^l = \gamma\omega_{ij}(\delta^{ik}n^j n^l)e_{kl} \\ &= \frac{1}{2}\left(\frac{\gamma}{2}\left(\delta^{ik}n^j n^l - \delta^{jl}n^i n^k\right)\right)e_{ij}e_{kl} \end{aligned}$$

Finally, The stiffness tensor a^{ijkl} is finally expressed as follows.

$$\begin{aligned}
a^{ijkl} = & An^i n^j n^k n^l \\
& + B(n^i n^j P^{kl} + P^{ij} n^k n^l) \\
& + C(P^{ij} P^{kl}) \\
& + D(P^{ik} P^{jl} + P^{il} P^{jk} - P^{ij} P^{kl}) \\
& + L(P^{ik} n^j n^l + P^{il} n^j n^k + P^{jk} n^i n^l + P^{jl} n^i n^k) \\
& + \frac{1}{4}(\alpha_{\perp} P_{mn} + \alpha_{\parallel} n_m n_n) \varepsilon^{mij} \varepsilon^{nkl} \\
& + \frac{\gamma}{2}(\delta^{ik} n^j n^l - \delta^{il} n^j n^k)
\end{aligned} \tag{9}$$

Setting the symmetry axis n along e_1 , i.e., the x -axis, each coefficient is expressed as follows.

normal block

$$\begin{aligned}
a^{1111} &= A, \\
a^{1122} &= a^{1133} = a^{2211} = a^{3311} = B, \\
a^{2222} &= a^{3333} = C + D, \\
a^{2233} &= a^{3322} = C - D.
\end{aligned}$$

(23) shear pair

$$\begin{aligned}
a^{2323} &= a^{3232} = D + \frac{\alpha_{\parallel}}{4}, \\
a^{2332} &= a^{3223} = D - \frac{\alpha_{\parallel}}{4}.
\end{aligned}$$

(12) shear pair

$$\begin{aligned}
a^{1212} &= L + \frac{\alpha_{\perp}}{4} - \frac{\gamma}{2}, \\
a^{1221} &= a^{2112} = L - \frac{\alpha_{\perp}}{4}, \\
a^{2121} &= L + \frac{\alpha_{\perp}}{4} + \frac{\gamma}{2}.
\end{aligned}$$

(13) shear pair

$$\begin{aligned}
a^{1313} &= L + \frac{\alpha_{\perp}}{4} - \frac{\gamma}{2}, \\
a^{1331} &= a^{3113} = L - \frac{\alpha_{\perp}}{4}, \\
a^{3131} &= L + \frac{\alpha_{\perp}}{4} + \frac{\gamma}{2}.
\end{aligned}$$

2.3 Effective medium of PML

Let us consider the case in which a PML is applied in the $+x$ -direction. In this case, based on Eq.(3) in Section 1, the stiffness tensor of the PML region can be expressed by interpreting the transformed domain as an effective medium. Since we assume that the PML is applied only along the $+x$ -axis, the Jacobian matrix Λ takes the form $\text{diag}(a, 1, 1)$.

According to the study of Chang et al. (2014), it has been shown that, by applying the polar decomposition to the Jacobian matrix and using the eigenvalues of the resulting symmetric tensor V , the transformed stiffness tensor can be expressed in the form given below. [1]

$$\tilde{C}^{IJKL} = \frac{1}{J} \lambda_I \lambda_K C^{IJKL} \quad (10)$$

where a capital index in the subscript takes the same value as its corresponding lower index, and $J = \prod_{I=1}^n \lambda_I$.

Restricting attention to the components relevant to the present discussion, we obtain

$$\begin{aligned} \tilde{C}^{1111} &= a(\lambda + 2\mu), \\ \tilde{C}^{1122} &= \tilde{C}^{1133} = \tilde{C}^{2211} = \tilde{C}^{3311} = \lambda, \\ \tilde{C}^{2222} &= \tilde{C}^{3333} = \frac{\lambda + 2\mu}{a}, \\ \tilde{C}^{2233} &= \tilde{C}^{3322} = \frac{\lambda}{a}, \\ \tilde{C}^{1212} &= a\mu, \quad \tilde{C}^{1221} = \tilde{C}^{2112} = \mu, \quad \tilde{C}^{2121} = \frac{\mu}{a}, \\ \tilde{C}^{1313} &= a\mu, \quad \tilde{C}^{1331} = \tilde{C}^{3113} = \mu, \quad \tilde{C}^{3131} = \frac{\mu}{a}, \\ \tilde{C}^{2323} &= \tilde{C}^{2332} = \tilde{C}^{3223} = \tilde{C}^{3232} = \frac{\mu}{a}. \end{aligned}$$

By matching this expression to the parameters of the transversely isotropic polar stiffness tensor derived above, each parameter can be written as follows.

$$\begin{aligned}
A &= a(\lambda + 2\mu), \\
B &= \lambda, \\
C &= \frac{\lambda + \mu}{a}, \\
D &= \frac{\mu}{a}, \\
L &= \frac{\mu}{4} \left(a + 2 + \frac{1}{a} \right) = \mu \frac{(a+1)^2}{4a}, \\
\alpha_{\parallel} &= 0, \\
\alpha_{\perp} &= \mu \left(a + \frac{1}{a} - 2 \right) = \mu \frac{(a-1)^2}{a}, \\
\gamma &= \mu \left(\frac{1}{a} - a \right).
\end{aligned}$$

2.4 Energy Analysis

Revisiting the strain energy density expression derived earlier, the case $n = e_1$ can be written as follows.

$$\begin{aligned}
\psi &= \frac{1}{2} A \varepsilon_{11}^2 + B \varepsilon_{11} \varepsilon_{\perp} + \frac{1}{2} C \varepsilon_{\perp}^2 + D E_{\perp} : E_{\perp} + 2L(\varepsilon_{12}^2 + \varepsilon_{13}^2) \\
&\quad + \frac{1}{2} \alpha_{\perp} (\omega_2^2 + \omega_3^2) + \frac{1}{2} \alpha_{\parallel} \omega_1^2 + \gamma (\varepsilon_{12} \omega_3 - \varepsilon_{13} \omega_2)
\end{aligned} \tag{11}$$

As derived above, there are no coupling terms between deformation components associated with distinct physical responses. Therefore, even when a particular type of deformation is applied, it does not induce other types of stress or energy contributions. On this basis, the total strain energy expression can be decomposed into four independently additive blocks that can be analyzed without mutual interference.

$$\psi_{\text{block}} = \begin{cases} \text{(i) Normal block} & \frac{1}{2} \begin{bmatrix} \varepsilon_{11} & \varepsilon_{\perp} \end{bmatrix} \begin{bmatrix} A & B \\ B & C \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{\perp} \end{bmatrix} \\ \text{(ii) Transverse deviatoric block} & D E_{\perp} : E_{\perp} \\ \text{(iii) Axial rotation block} & \frac{1}{2} \alpha_{\parallel} \omega_1^2 \\ \text{(iv) Shear-rotation block} & \frac{1}{2} \begin{bmatrix} \varepsilon & \omega \end{bmatrix} \begin{bmatrix} 4L & \pm\gamma \\ \pm\gamma & \alpha_{\perp} \end{bmatrix} \begin{bmatrix} \varepsilon \\ \omega \end{bmatrix} \end{cases} \tag{12}$$

Since the PML stretch factor is complex, we investigate the conditions on the complex parameters under which each block matrix is Hermitian positive semidefinite. First, let us define

$$a = \alpha + i \frac{\beta}{\omega}.$$

Substituting this into the previously derived parameters and taking the real part yields the following expressions.

$$\Re A = (\lambda + 2\mu)\alpha$$

$$\Re B = \lambda$$

$$\Re C = (\lambda + \mu) \frac{\alpha}{|a|^2}$$

$$\Re D = \mu \frac{\alpha}{|a|^2}$$

$$\Re L = \frac{\mu}{4} \left(\alpha + 2 + \frac{x}{|a|^2} \right)$$

$$\Re \alpha_{\parallel} = 0$$

$$\Re \alpha_{\perp} = \mu \left(\alpha + \frac{\alpha}{|a|^2} - 2 \right)$$

$$\Re \gamma = \mu \left(\frac{\alpha}{|a|^2} - \alpha \right).$$

We now derive the conditions under which each block satisfies the Hermitian positive semidefinite condition.

(i) Normal block

$$\frac{1}{2} \begin{bmatrix} \varepsilon_{11} & \varepsilon_{\perp} \end{bmatrix} \begin{bmatrix} A & B \\ B & C \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{\perp} \end{bmatrix}$$

For complex coefficients, the Hermitian part is

$$H_N = \begin{bmatrix} \Re A & B \\ B & \Re C \end{bmatrix},$$

so the positive semidefinite conditions are

$$\Re A \geq 0, \quad \Re C \geq 0, \quad (\Re A)(\Re C) - B^2 \geq 0.$$

Substituting these yields

$$(\lambda + 2\mu)\alpha \geq 0$$

$$(\lambda + \mu) \frac{\alpha}{|a|^2} \geq 0$$

$$(\lambda + 2\mu)(\lambda + \mu) \frac{\alpha^2}{|a|^2} - \lambda^2 \geq 0.$$

Rearranging the last equation yields

$$\mu(3\lambda + 2\mu)\alpha^2 - \lambda^2 \frac{\beta^2}{\omega^2} \geq 0.$$

Thus, the conditions for the normal block are

$$\boxed{\alpha \geq 0, \quad \mu(3\lambda + 2\mu)\alpha^2 \geq \lambda^2 \frac{\beta^2}{\omega^2}.} \quad (13)$$

(ii) Transverse deviatoric block

For this block, we only need to consider D .

Condition:

$$\Re D \geq 0$$

That is,

$$\mu \frac{\alpha}{|a|^2} \geq 0.$$

If $\mu > 0$ in the background medium, we have

$$\boxed{\alpha \geq 0} \quad (14)$$

(iii) Axial rotation block

It is given by

$$\frac{1}{2}\alpha_{\parallel}\omega_1^2$$

but in this case,

$$\alpha_{\parallel} = 0.$$

Therefore,

$$\boxed{\alpha_{\parallel} = 0} \quad (15)$$

(iv) Shear-rotation block

For example, the $(12)\text{-}\omega_3$ block is

$$2L\varepsilon_{12}^2 + \gamma\varepsilon_{12}\omega_3 + \frac{1}{2}\alpha_{\perp}\omega_3^2,$$

and the Hermitian part is

$$H_S = \begin{bmatrix} 4\Re L & \Re \gamma \\ \Re \gamma & \Re \alpha_{\perp} \end{bmatrix}.$$

The positive semidefinite conditions are

$$4\Re L \geq 0, \quad \Re \alpha_{\perp} \geq 0, \quad (4\Re L)(\Re \alpha_{\perp}) - (\Re \gamma)^2 \geq 0.$$

Now, substituting these yields

$$4\Re L = \mu \left(\alpha + 2 + \frac{\alpha}{|a|^2} \right),$$

$$\Re \alpha_{\perp} = \mu \left(\alpha + \frac{\alpha}{|a|^2} - 2 \right),$$

$$\Re \gamma = \mu \left(\frac{\alpha}{|a|^2} - \alpha \right).$$

Thus, the determinant is

$$\begin{aligned} & (4\Re L)(\Re \alpha_{\perp}) - (\Re \gamma)^2 \\ &= \mu^2 \left(\alpha + 2 + \frac{\alpha}{|a|^2} \right) \left(\alpha + \frac{\alpha}{|a|^2} - 2 \right) - \mu^2 \left(\frac{\alpha}{|a|^2} - \alpha \right)^2. \end{aligned}$$

Expanding this yields a very neat expression:

$$\boxed{(4\Re L)(\Re \alpha_{\perp}) - (\Re \gamma)^2 = -\frac{4\mu^2 \beta^2}{\omega^2 |a|^2} \leq 0} \quad (16)$$

As can be seen from Eq.(16), the determinant of the shear-rotation block is always less than or equal to zero. This implies that, when a is complex, the Hermitian positive semidefinite condition cannot in general be satisfied. In conclusion, due to the violation of the non-negativity of the energy term associated with the coupling between out-of-plane shear and rotation, the effective medium induced by the PML admits the possibility of pointwise energy divergence.

References

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